

WASP-43b

R-2530

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140407 · created 2026-08-25T23:04:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456754.77285 +/- 0.00069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.014
Transit depth (Rp/Rs)^2	2.59 +/- 0.46 [%]
Orbital Inclination (inc)	83.29 +/- 1.53
Airmass coefficient 1 (a1)	1.232 +/- 0.013
Airmass coefficient 2 (a2)	-0.1156 +/- 0.006
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.14
Optimal Annulus	9.37
Transit Duration (day)	0.0537 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.014 vs 0.1594 (deviation 0.11σ)
Duration fit vs published	0.0537 d vs 0.0512 d (deviation 0.58σ)
Mid-time O–C	-1.9 min
Depth SNR	11.5
Residual scatter	1.07 %

Light curve

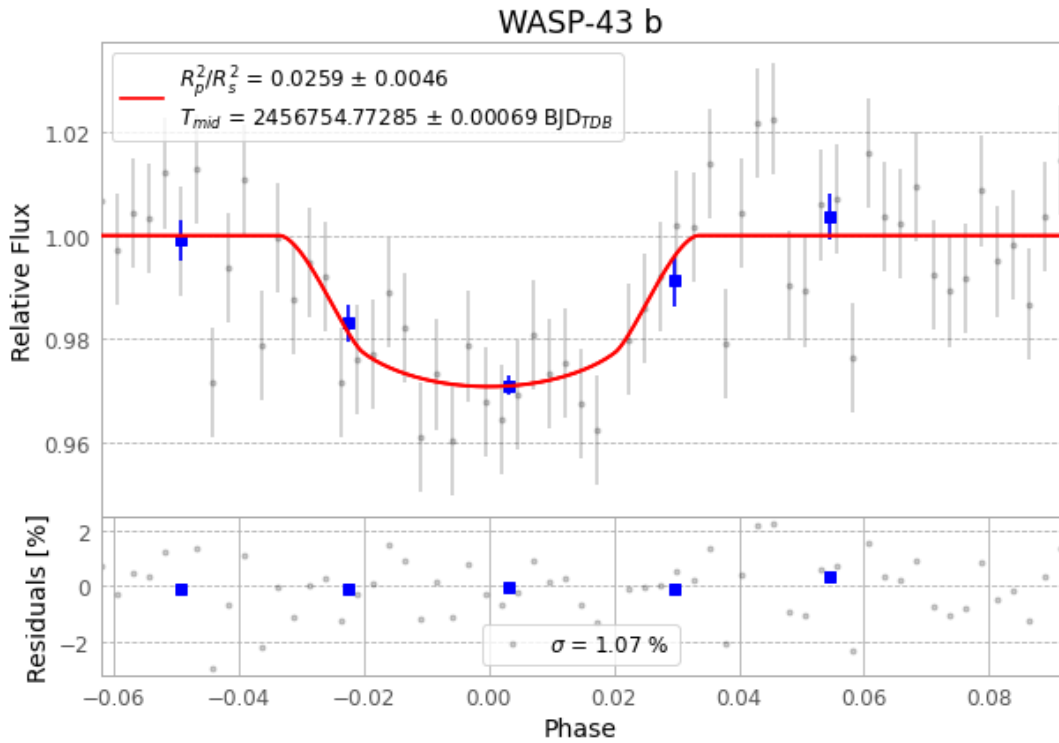


Figure 1 — FinalLightCurve_WASP-43 b_2014-04-06.png (SHA-256 9320cd104782d302...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 239], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 296e5ac52aefa627...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-43140407051213.FITS → WASP-43140407081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140407.log (ef31f8df3b63...), FinalLightCurve_WASP-43 b_2014-04-06.png (9320cd104782...), FinalLightCurve_WASP-43 b_2014-04-06.csv (961f86516e18...)

Compute resources

Wall time 102.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)